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Triple-isotopic analyses pinpoint microbial methane release from subsea permafrost in the inner Laptev Sea



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Elevated methane concentrations are observed in the shallow water column above subsea permafrost on the East Siberian Arctic Shelf, including the inner Laptev Sea. The subsea source of this methane is poorly understood, yet crucial for predicting future methane emissions. Here, we combine analyses of dissolved methane concentrations, triple-isotopic fingerprinting, and Bayesian Markov Chain Monte Carlo statistics to constrain the source of high-concentration methane sampled during four expeditions, 2016–2020. In contrast to earlier findings of predominantly thermogenic methane release in the outer Laptev Sea, our results for the major methane release region of the inner Laptev Sea suggest that old microbial methane (radiocarbon age >48000 y BP) is being released from preformed methane pools stored within the subsea permafrost system. This observation reveals that several sources contribute to the elevated methane on the shelf and thus the necessity to consider a diversity of sources when estimating future methane release trajectories.

Highly elevated levels of methane (CH₄) have been detected across extensive scales of the East Siberian Arctic Shelf (ESAS) over the past two decades^{1–3}. Next to enriched CH₄ concentrations in both seawater and the atmospheric boundary layer, strong ebullition was observed at several seep sites across the ESAS^{3–5}. The ESAS (<200 m depth) is the world's largest shelf sea (~1.7 × 10⁶ km²), yet it is very shallow with an average depth of only ~45 m⁶. It started forming ~15 ky before present (BP) as the sea level rose, inundating Pleistocene-age terrestrial permafrost^{7–10}. The resulting subsea permafrost below the ESAS is estimated to stretch over ~1.5 × 10⁶ km² at an average thickness of ~330 m^{11,12}. Based on subsea drill coring through the upper horizon of the subsea permafrost, repeated 21–22 years apart, it was shown that subsea permafrost is now thawing ~35 times faster than nearby terrestrial permafrost^{10,13}. The thawing subsea permafrost contains preformed CH₄ and organic matter (OM) that could contribute to the observed intense CH₄ emissions. However, the CH₄ sources releasing CH₄ to the ESAS seawater are poorly constrained. Constraining the relative importance of the different CH₄ sources is a prerequisite for predicting to what extent the observed CH₄ emissions in the ESAS will intensify in the future.

Several old subsea pools could contribute to the elevated CH₄ levels on the ESAS. The deepest subsea CH₄ sources are fossil natural gas reservoirs of

either microbial or thermogenic origin, formed below the permafrost layer^{14,15}. The amount of CH₄ stored in such reservoirs is uncertain but may be ~1–5 GtC for the ESAS^{16–18}, i.e., respectively ~3–21% and ~1–7% of Russia's natural gas reserves and resources¹⁹. The overlying subsea permafrost also holds substantial amounts of carbon in the form of both preformed CH₄ and as OM that can be microbially degraded to CH₄ upon thaw^{13,20}. The preformed CH₄ within the subsea permafrost layer could originate from past in situ methanogenesis^{21,22}. Furthermore, upward-moving gases from sub-permafrost fossil gas reservoirs could get trapped within²³ or below¹⁴ the subsea permafrost layer.

The amounts of both OM and CH₄ in subsea permafrost reservoirs are highly uncertain. Estimates of the amount of OM in subsea permafrost are ~2800 GtC for the circum-Arctic²⁴ and ~500 GtC for the ESAS². The CH₄ trapped by subsea permafrost may be present as free gas but also as gas hydrate, a solid where water (H₂O) and CH₄ molecules combine to a crystalline structure²⁵. A gas hydrate is stable within specific pressure-temperature boundaries, where higher pressures and lower temperatures favor the hydrate phase²⁶. An overview of estimated amounts of preformed CH₄ stored in oceanic and (subsea) permafrost hydrates, as free gas/hydrate within subsea permafrost, and in gas reserves/resources is provided in Supplementary Table 1, along with a description of how they were

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estimated, as far as that is retrievable. The highly varying estimates of the carbon mass of preformed CH₄ stored within the ESAS subsea permafrost domain range in published studies from 1²⁷ to 900 GtC². Taken together, there are very large uncertainties in the size of CH₄-relevant compartments in the Arctic sedimentary drape.

There are multiple mechanisms that can release CH₄ from the old subsea CH₄ and OM reservoirs. First, the combination of geothermal heat, warm overlying Holocene seawater, and human-induced additional climate warming results in subsea permafrost thawing^{1,4,10}. Such warming may induce microbial degradation of OM and concomitant CH₄ production in the thawed subsea permafrost, but also results in the direct release of subsea permafrost-trapped CH₄ by the destabilization of the subsea permafrost itself^{2,28–30}. Existing and developing taliks, i.e., unfrozen layers/bodies in permafrost³¹, can act as migration pathways for the release of subsea permafrost-trapped CH₄ to the water column^{2,10,29,32,33}. Furthermore, increasing subsea temperatures could destabilize CH₄ hydrates²⁶, releasing CH₄ while facilitating the upward transport of formerly trapped preformed CH₄ from greater depths. Finally, the ESAS is tectonically active^{34–37}, and CH₄ originating from fossil gas reservoirs could seep along faults formed by tectonic activity^{2,28,38,39}.

In addition to these old subsea CH₄ sources, several younger and/or laterally transported sources can potentially also contribute CH₄ to the water column. Holocene shelf sediments contain both laterally transported OM (from coastal erosion or river discharge) and marine OM, which can be microbially degraded to CH₄ under anoxic conditions. Rivers^{32,40} and potentially submarine groundwater discharge⁴¹ can transport dissolved CH₄ to the ESAS, although dissolved CH₄ is largely reduced in concentration during freshwater transport to the sea⁴².

The heterogeneity of the subsea geology and permafrost in the ESAS precludes generalizing CH₄ emissions to the water column and the atmosphere based on the currently available data. While the origin of high CH₄ concentrations at outer Laptev Sea hotspot locations was identified as predominantly stemming from fossil thermogenic gas³, the origin of CH₄ released in other regions of the ESAS remains largely unknown. Specifically, the inner Laptev Sea has been found to hold extensive seeps^{4,10} and highly elevated CH₄ concentrations^{1,2,4}, yet CH₄ sources are poorly constrained. Given its shallow depth (<25 m), there is a short conduit to the atmosphere for any sediment-escaping CH₄.

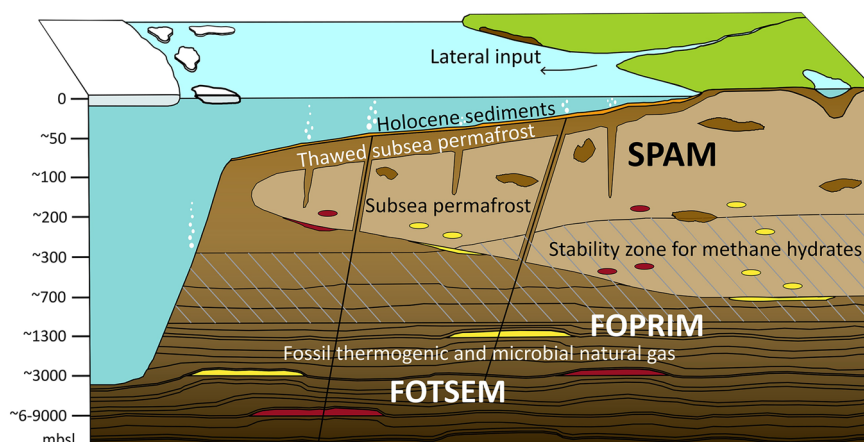
Several traits make permafrost underneath the inner Laptev Sea susceptible to destabilization. The region has been inundated comparatively recently (starting ~10 ky BP^{7,8}) and is suggested to have relicts of its terrestrial past in the form of taliks from paleo-rivers and thermokarst processes^{10,11,29,43–45}. This coastal sea also has more pronounced lateral inputs than deeper offshore regions, including heat from the Lena River plume³². Moreover, the few existing core-based measurements of shallow coastal subsea permafrost indicate a thaw rate ~35 times greater than that of its terrestrial counterpart^{10,13} and, in the upper 57 m, a temperature 8–11 °C higher⁴. This suggests that the shallow subsea permafrost is much more

vulnerable than nearby terrestrial permafrost, likely because of the long-term heating from overlying seawater^{29,46}. The geological setting, highly elevated CH₄ levels, and vulnerability of the inner Laptev Sea subsea permafrost make this a key subregion to disentangle what sources are dominating the CH₄ releases in the ESAS.

The rapid and amplified warming of the Arctic additionally motivates the study of sources and release mechanisms of CH₄ in the inner Laptev Sea. The Arctic is warming three to four times faster than the global average^{47,48}, and a warmer climate would likely accelerate CH₄ release in the ESAS region. Benthic seawater temperatures in the shallow ESAS (<25 m) have increased substantially compared to those in the deeper parts of the shelf²⁹. The observational record shows that mean summer bottom-water temperatures increased by ~1 °C for the 1999–2009 decade compared to the 100-year mean^{4,29}, a process which may have accelerated over the past decade concomitant with wider Arctic warming trends. Potential effects of rising temperatures are increased CH₄ hydrate destabilization, increased subsea permafrost destabilization, releasing subsea permafrost-trapped CH₄, increased OM degradation in thawing subsea permafrost, and increased lateral inputs of OM and CH₄. Some of these positive feedback mechanisms to climate change may evolve gradually, while others may be more abrupt. Future CH₄ emissions from thawing (subsea) permafrost and destabilizing CH₄ hydrates are two of the large uncertainties in the changing CH₄ budget and are considered potential biogeochemical tipping points for climate change^{49,50}. In addition, the estimated size of the subsea OM and CH₄ deposits in the ESAS is ~1–4 × 10² times larger than the total amount of CH₄ in the atmosphere (3.8 GtC)^{2,27,50}. However, the current atmospheric monitoring networks cannot detect a sudden multiple-year-long CH₄ release event in the ESAS of the size of a 100% increased oceanic flux⁵¹. Improved constraints on the type and evolution of CH₄ emissions in the ESAS are necessary for reliable projections of this region's current and future, potentially intensifying CH₄ emissions.

The main objective of this study is to quantify the relative contribution of different sources to the observed elevated dissolved CH₄ in the seawater of the inner Laptev Sea. To that end, we measured the concentration and triple-isotopic ($\Delta^{14}\text{C}$, $\delta^{13}\text{C}$, $\delta^2\text{H}$) composition of seawater CH₄ collected in regions at and near the inner Laptev Sea CH₄ hotspots over multiple years. The concentration of CH₄ and its stable isotopic composition were measured in samples from four ship expeditions between 2016 and 2020 (121 samples for $\delta^{13}\text{C}_{\text{CH}_4}$, 76 for $\delta^2\text{H}_{\text{CH}_4}$), and the natural abundance radiocarbon content in samples from 2019 ($n = 7$) and 2020 ($n = 16$). The $\delta^{13}\text{C}_{\text{CH}_4}$ and $\delta^2\text{H}_{\text{CH}_4}$ data were evaluated to characterize the formation process of CH₄ and distinguish between primary microbial and thermogenic CH₄^{52–54}. Radiocarbon measurements were used to determine the age of the CH₄ precursor^{3,55}. These observations were interpreted within a comprehensive literature review to constrain the isotopic composition of possible CH₄ sources (Fig. 1). These constrained endmembers and the measured CH₄ isotope data were combined in a Bayesian Markov Chain Monte Carlo (MCMC) statistical

Fig. 1 | Methane sources in the East Siberian Arctic Shelf subsea system and the three main old (>48 ky BP) endmembers. The endmember subsea permafrost-associated methane (SPAM) is formed/situated in the (thawing) subsea permafrost layer. The endmembers fossil primary microbial (FOPRIM) gas (marked with yellow) and fossil thermogenic and secondary microbial (FOTSEM) gas (marked with red) are present in preformed CH₄ pools. The vertical scale indicates approximate depths in meters below sea level (mbsl). Detailed descriptions and stable isotope compositions of CH₄ from these endmember compartments are provided in Supplementary Methods 1.2 and Supplementary Data 1.



framework^{56,57} to quantify the dominant sources of dissolved CH₄ in the inner Laptev Sea.

Results and discussion

Elevated methane concentrations and ubiquitous bubble plumes in the inner Laptev Sea

High CH₄ concentrations and many bubble plumes were observed in the inner Laptev Sea during the four expeditions. A major hotspot region with strongly elevated CH₄ concentrations (up to 27,000 nM; Supplementary Data 2–5) was repeatedly observed along with dense ebullition patterns (Fig. 2; Supplementary Note 3 and Supplementary Data 6). North and south of this major methane hotspot (MMH), CH₄ concentrations were still up to 150 nM for samples retrieved during 2016–2020; all of these are substantially elevated compared to ~4 nM that would result if there were no local source and just equilibration with atmospheric CH₄.

There is some variation between years in spatial concentration gradients and in the locations of observed CH₄ hotspots. The MMH observed in 2016–2020 was slightly northward compared to CH₄ hotspots identified earlier. In 2005, bottom water samples showed a distinct CH₄ hotspot 101 km away from the recent MMH (Fig. 2)². In 2007, CH₄ hotspots beneath sea ice were observed 79–57 km away from this MMH¹. In 2009 and 2010, water column data showed a major CH₄ hotspot about 68 km from the MMH⁴. Varying locations of the most apparent CH₄ hotspots over 15 years indicate that multiple CH₄ hotspots are present in the inner Laptev Sea and possibly have a changing activity level over short time scales (years to

decades). The central impression is that there is a continuous ubiquity of intensive CH₄ hotspots in this inner Laptev Sea region.

At the MMH, dissolved CH₄ concentrations were up to 6000 times higher than expected from equilibrium with the atmosphere. The highest CH₄ concentrations were observed below the pycnocline (~10–15 m deep) in the bottom water layer (Fig. 2 and Supplementary Note 2). The geographical alignment of the highest CH₄ concentrations with the intensity of the ebullition pattern at the MMH suggests partial dissolution from rising bubbles as a key cause of the elevated seawater CH₄ concentrations. These combined findings point to one or several continuous seabed sources of CH₄ causing the perennial and strongly elevated CH₄ concentrations in the shallow water column of the inner Laptev Sea.

Radiocarbon signatures of dissolved methane

Radiocarbon dating of dissolved CH₄ in the inner Laptev Sea points toward the release of old (>48 ky BP) CH₄ into the overlying seawater, and mixing of this primary source with a background of younger CH₄. Keeling plots were used to identify the isotopic signature of the primary CH₄ source, i.e., the y-intercept of the Keeling plot (“Methods”). The Keeling plot for the combined 2019 and 2020 measurements of $\Delta^{14}\text{C}_{\text{CH}_4}$ yielded a source signature of $-1058 \pm 66\text{‰}$ (95% confidence interval), with a closely corresponding signature for 2020 data at the MMH of $-1007 \pm 8\text{‰}$ (Fig. 3). As the lowest $\Delta^{14}\text{C}$ limit for radiocarbon dating by AMS was -998‰ , the reported source signatures point to a $^{14}\text{C}_{\text{age}} > 48 \text{ ky BP}$ (calculations in Supplementary Note 4). Hence, we conclude that the strong CH₄ releases are

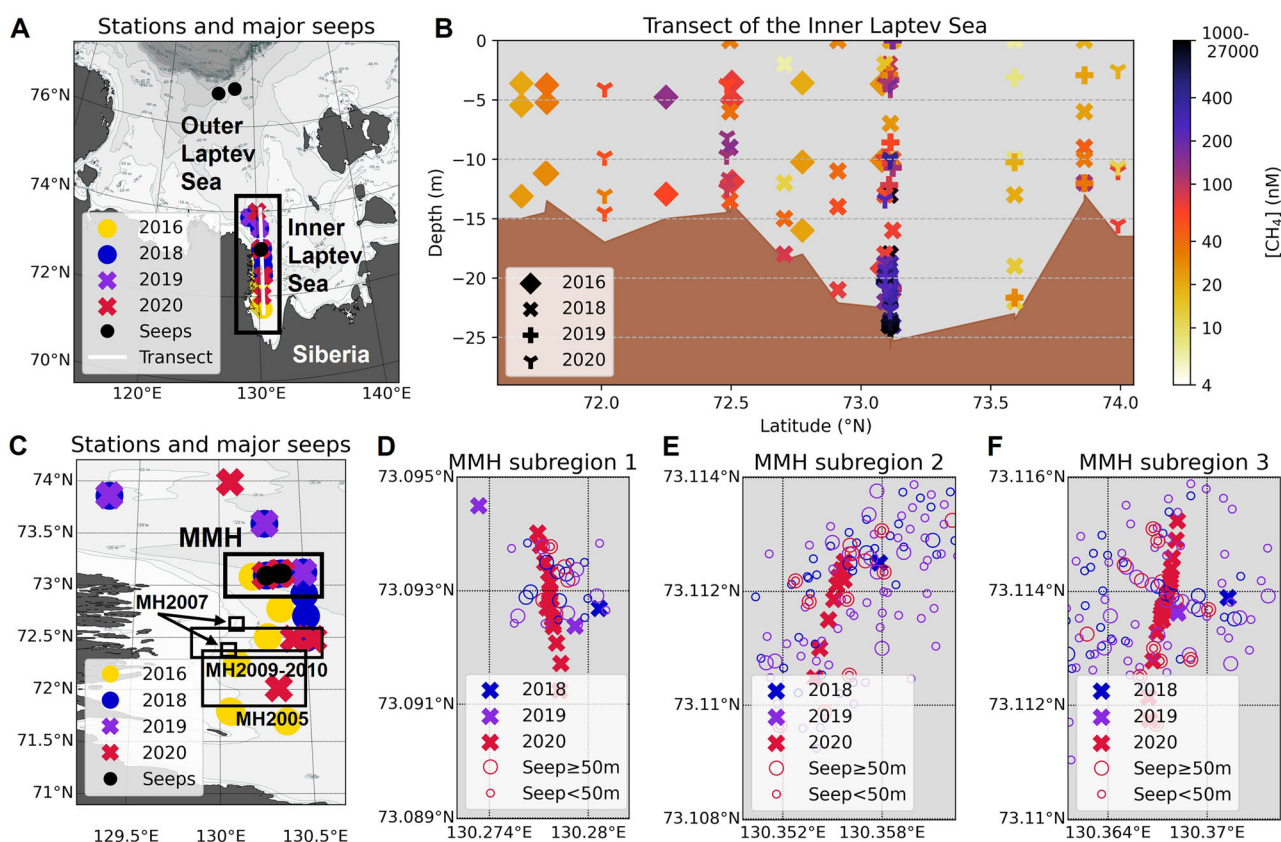


Fig. 2 | Sampling stations, major seeps, and observed CH₄ concentrations in the inner Laptev Sea. **A** The sampling stations of this study and the major seeps in the inner Laptev Sea (this study) and the outer Laptev Sea³. The different expedition years are marked with the following colors: 2016 (yellow), 2018 (dark blue), 2019 (purple), 2020 (red). **B** A transect of the inner Laptev Sea showing multi-year CH₄ concentrations. Samples of each expedition year are shown with different markers. Concentration profiles for each year separately are in Supplementary Note 2. The shown bottom depths are based on conductivity, temperature, and depth (CTD) data of each sampling station. **C** The inner Laptev Sea study area with multi-year

sampling stations, the major seeps, the major methane hotspot (MMH), and formerly observed methane hotspots in 2005 (MH2005)², 2007 (MH2007)¹, and 2009–2010 (MH2009–2010)⁴. **D–F** The sampling locations and detected seeps (seep size smaller than or larger than/equal to 50 m) for the different expedition years. The observed seeps in **D–F** stretch further beyond the shown subregions. This adds to bubble flares that have been reported for these regions and the wider inner Laptev Sea region earlier⁴. Sample details, measured CH₄ concentrations, seep sizes, and CTD data are given in Supplementary Data 2–10.

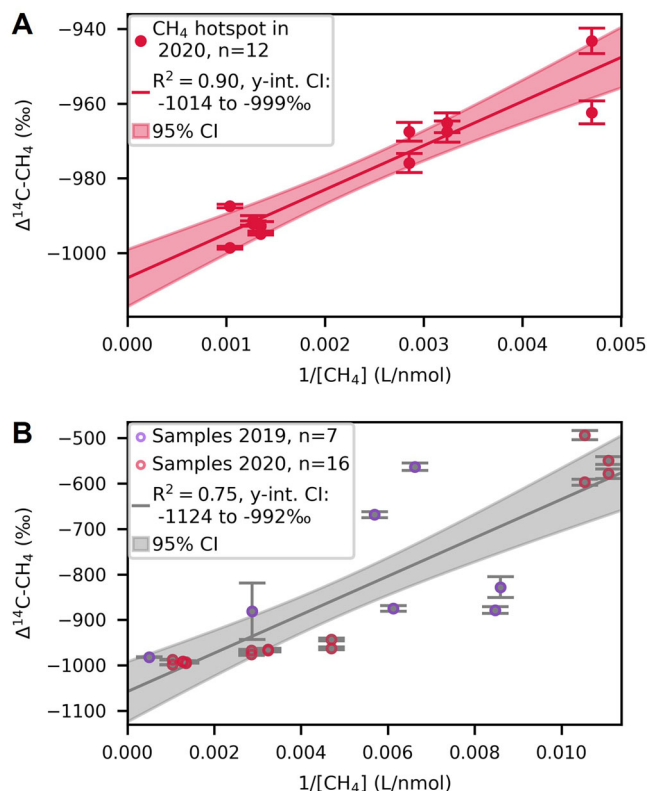


Fig. 3 | Keeling plots of inner Laptev Sea $\Delta^{14}\text{C}_{\text{CH}_4}$ data. **A** Keeling plot of $\Delta^{14}\text{C}_{\text{CH}_4}$ data at the major CH_4 hotspot for 2020 (in red). **B** Keeling plot of $\Delta^{14}\text{C}_{\text{CH}_4}$ data at and near the major CH_4 hotspot (72.4–73.2°N) for 2019 and 2020 combined (in gray). Samples of 2019 and 2020 are marked with purple and red circles, respectively. The shaded areas indicate the 95% confidence interval (CI), and the error bars show ± 2 times the standard uncertainty of each data point. The y-intercept and its 95% CI represent the main source signature. Complete sample lists and radiocarbon data are given in Supplementary Data 4 and 5.

from a very old ($^{14}\text{C}_{\text{age}} > 48$ ky BP) CH_4 source or sources. Therefore, only compartments with such ancient radiocarbon signatures need to be considered as endmembers to apportion the dominant CH_4 sources in the inner Laptev Sea.

This old CH_4 could originate from a preformed CH_4 pool below or within the subsea permafrost or represent recently produced CH_4 from deep-situated thawing permafrost having radiocarbon-depleted OM. Three old source endmembers were consistent with this picture (Fig. 1 and “Methods”, Supplementary Methods 1.2). The first endmember was subsea permafrost-associated methane (SPAM), representing microbial CH_4 formed from OM in the upper subsea permafrost region before/during freezing and/or formed upon subsea permafrost thawing. The endmember SPAM thus represents CH_4 that is freshly formed near the subsea permafrost thaw front, as well as CH_4 formed in the past and retained in the current upper subsea permafrost layer (Fig. 1). The second endmember was fossil primary microbial (FOPRIM) gas, being primary microbial gas formed in the past by OM degradation using older OM than SPAM at higher formation temperatures. The third endmember was fossil thermogenic and secondary microbial (FOTSEM) gas, representing CH_4 formed by catagenesis in even deeper sedimentary basins at even higher temperatures and secondary microbial gas formed by the biodegradation of thermogenic petroleum. The FOPRIM and FOTSEM gases represent preformed CH_4 pools that may be located at multiple locations: in gas reservoirs beneath the subsea permafrost, or trapped below/within the permafrost as either free gas or CH_4 hydrate (indicated by the yellow and red areas in Fig. 1). Details regarding these endmembers are in Supplementary Methods 1.2.

The release of CH_4 from ancient seabed sources to the bottom seawater has been proposed for two other Arctic shelf sea regions; however, in both cases, the generated isotopic data were limited and based on single expeditions. The reports for the Beaufort Sea did not feature hydrogen isotope data and did not utilize the generated stable carbon isotope data for more detailed source apportionment⁵⁵. The identified ancient CH_4 was suggested to originate from CH_4 hydrates, permafrost, and/or geologic sources⁵⁵. For the outer Laptev Sea, very few $\Delta^{14}\text{C}_{\text{CH}_4}$ data were generated³. These were combined with single-year dual-stable isotope data, and the release of thermogenic CH_4 was suggested³.

In addition to constraining the old characteristics of the strong CH_4 emission source, the $\Delta^{14}\text{C}_{\text{CH}_4}$ data in the Keeling plots also yield information on the character/source of the CH_4 pool of lower concentration that it is mixing into. The consistently higher $\Delta^{14}\text{C}_{\text{CH}_4}$ values associated with lower CH_4 concentrations observed here indicate a background of less-aged CH_4 , which appears to be relatively constant over 2019–2020, considering the good fit in the Keeling plot for the years combined (Fig. 3B, $R^2 = 0.75$). The relatively younger background pool mixing with large amounts of old CH_4 at the MMH could originate from CH_4 production in the upper reaches of thawed subsea permafrost and/or in sediment layers accumulated during the Holocene, or preformed CH_4 transported by rivers. Any of these younger CH_4 sources deliver microbially produced CH_4 .

The lowest concentration samples analyzed for radiocarbon content were 90–95 nM and had an average $\Delta^{14}\text{C}_{\text{CH}_4}$ ($\pm 1\sigma$) value of $-555 \pm 45\%$ ($n = 4$, Supplementary Data 5); these samples are expected to have the highest relative contribution of background CH_4 among the sample set. River input is, in principle, a potentially contributing source of CH_4 . However, the dissolved CH_4 concentrations in the Lena Delta channels exporting CH_4 to the sea are only 5–20 nM⁴², so river input seems unlikely to be the primary cause of the observed elevated CH_4 concentrations of 90–95 nM. The younger CH_4 in the lower-concentration range of the Keeling mixing curves may stem from OM degradation in the upper range of thawed subsea permafrost and/or Holocene-accumulated sediments. The CH_4 concentration resulting from early diagenesis in the upper sediments was estimated by using a three-source mixing model (including CH_4 from atmospheric equilibrium, CH_4 produced by early diagenesis in surface sediments, and radiocarbon-free CH_4 from greater depths; see equations in Supplementary Note 5). In these computations, we used the $\Delta^{14}\text{C}$ value of upper sediment bulk organic carbon (OC) sampled in the region 130–131°E, 71–74°N^{58,59} to estimate the $\Delta^{14}\text{C}$ value of CH_4 produced by early diagenesis from shallow-sedimentary OM. If deeper and older sediments/thawed subsea permafrost, having lower $\Delta^{14}\text{C}$ values, would be degraded, a larger contribution from early diagenesis would be inferred. Furthermore, the reactivity of OM is compound-specific⁶⁰, and the sedimentary OM consists of a mixture of compounds from different origins and ages⁶¹. Therefore, bulk OC $\Delta^{14}\text{C}$ values are not directly equal to the $\Delta^{14}\text{C}$ of the organic compounds that are predominantly degraded to CH_4 . Consequently, the calculated CH_4 concentrations resulting from early diagenesis (68 ± 9 nM) should be considered a rough estimate. Taken together, the lower concentration background CH_4 may stem primarily from early diagenesis in Holocene sediments and/or the upper thawed subsea permafrost layer, contributing CH_4 to the seawater over some areas in the inner Laptev Sea, yet it represents a minor part of the highly elevated CH_4 observed at the MMH.

As $\Delta^{14}\text{C}$ measurements are, by definition, corrected for isotopic fractionation⁶² and thereby unaffected by microbial oxidation, the strong $\Delta^{14}\text{C}_{\text{CH}_4}$ Keeling regressions (Fig. 3) inform about the source signature. The good fits of the regression lines, combined with slopes being evidently larger than zero, support a scenario where a primary source mixes with a background of much lower CH_4 concentration. Microbial degradation of CH_4 in the sediment or water column would result in a regression slope close to zero. Hence, there is an overwhelmingly old source causing the highly elevated CH_4 concentrations, which is mixing with a background pool of younger CH_4 . This supports the use of Keeling plots to interpret the stable isotope data in a mixing scenario, as discussed next.

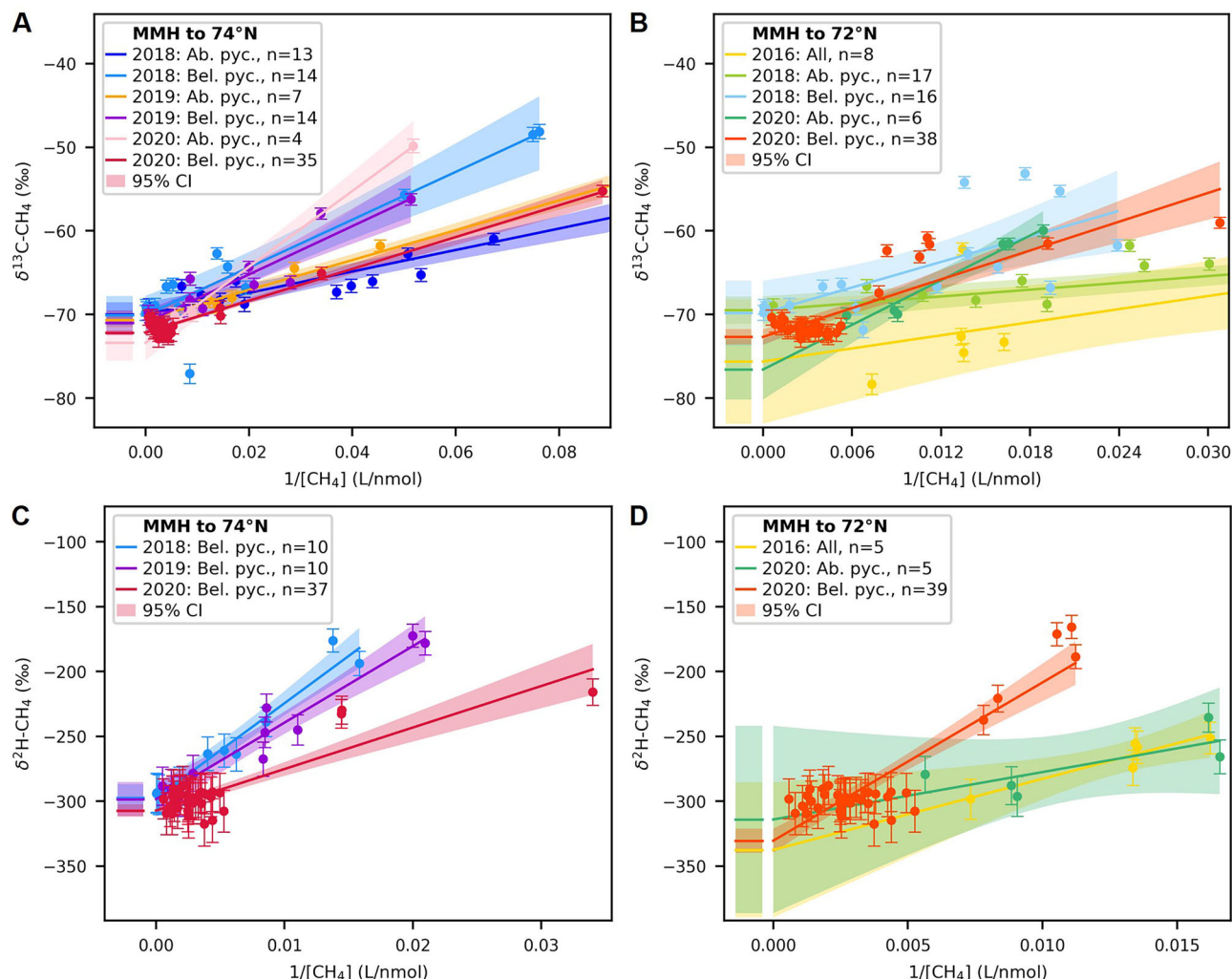


Fig. 4 | Multi-year $\delta^{13}\text{C}_{\text{CH}_4}$ and $\delta^2\text{H}_{\text{CH}_4}$ Keeling plots at and near the major CH_4 hotspot (MMH) in the inner Laptev Sea. Dual-stable isotope Keeling plots are indicated with different colors for different years, regions, and for both below pycnocline (Bel. pyc.) and above pycnocline (Ab. pyc.) samples: 2016 (yellow), 2018, 73.0–74.0°N, Ab. pyc. (dark blue), Bel. Pyc. (light blue), 72.0–73.2°N, Ab. pyc. (light green), Bel. Pyc. (light blue), 2019, 73.0–74.0°N, Ab. pyc. (gold), Bel. Pyc. (purple), 2020, 73.0–74.0°N, Ab. pyc. (light pink), Bel. Pyc. (red), 72.0–73.2°N, Ab. pyc. (dark green), Bel. Pyc. (orange). The shown y-intercepts represent the derived source signatures. The shaded areas represent the 95% confidence interval (CI). Individual

isotopic measurements are plotted with error bars representing ± 2 times the standard uncertainty. The y-intercepts and stable isotope data are in Supplementary Table 2 and Supplementary Data 2–5. **A** Multi-year $\delta^{13}\text{C}_{\text{CH}_4}$ Keeling plots of measurements at and north of the MMH (73.0–74.0°N). **B** Multi-year $\delta^{13}\text{C}_{\text{CH}_4}$ Keeling plots of measurements at and south of the MMH (72.0–73.2°N). **C** Multi-year $\delta^2\text{H}_{\text{CH}_4}$ Keeling plots of measurements at and north of the MMH (73.0–74.0°N). **D** Multi-year $\delta^2\text{H}_{\text{CH}_4}$ Keeling plots of measurements at and south of the MMH (72.0–73.2°N).

Dual-stable isotope signatures of dissolved methane

Multi-year dual-stable isotope results of dissolved CH_4 in the inner Laptev Sea indicate that microbially-produced CH_4 is the primary source, and that the source signature is very similar for all four studied years. The Keeling plot-derived multi-year average source signatures ($\pm 1\sigma$) for the below pycnocline samples were $-72 \pm 2\text{‰}$ for $\delta^{13}\text{C}_{\text{CH}_4}$ and $-313 \pm 19\text{‰}$ for $\delta^2\text{H}_{\text{CH}_4}$ (Fig. 4 and Supplementary Table 2). The multi-year average source signature ($\pm 1\sigma$) using the above pycnocline samples was very similar at $-73 \pm 3\text{‰}$ for $\delta^{13}\text{C}_{\text{CH}_4}$, and -314‰ for $\delta^2\text{H}_{\text{CH}_4}$ in 2020 (Fig. 4 and Supplementary Table 2). These low stable carbon and hydrogen isotope values point to primary microbially formed CH_4 ^{53,54}. Depending on the pathway by which CH_4 is released to the water column (bubble dissolution versus diffusion from the sediment), CH_4 may be susceptible to anaerobic CH_4 oxidation during its sedimentary transit, altering its stable isotope composition⁵⁴. The observed dual-stable isotope source signature is, however, very constant between years (Fig. 4 and Supplementary Table 2) and ebullition is ubiquitously observed (Fig. 2). Therefore, bubble dissolution likely represents the dominant pathway for CH_4 addition to the water

column. Hence, anaerobic oxidation of CH_4 in the sediments would, if present, only marginally affect the observed dual-stable isotope source signatures.

The observed source signature for the inner Laptev Sea is in substantial contrast to observations for the outer Laptev Sea, where source signatures of -43‰ to -55‰ for $\delta^{13}\text{C}_{\text{CH}_4}$ and -137‰ to -158‰ for $\delta^2\text{H}_{\text{CH}_4}$ were identified, and consequently a thermogenic origin of CH_4 was suggested for that region ~350 km offshore³. Combining the stable isotope and $\delta^{14}\text{C}_{\text{CH}_4}$ data from the inner Laptev Sea indicates a predominantly old microbial source, in our source compartment terms SPAM and/or FOPRIM, being released into the inner Laptev Sea seawater at the MMH, which contrasts with the thermogenic FOTSEM seeps in the outer Laptev Sea.

Isotopic source apportionment of methane released in the inner Laptev Sea

The Keeling plot-extrapolated isotopic source signatures of all four years of data fall within the endmember region of SPAM (Fig. 5). This suggests that CH_4 produced microbially from old subsea permafrost OM is the primary

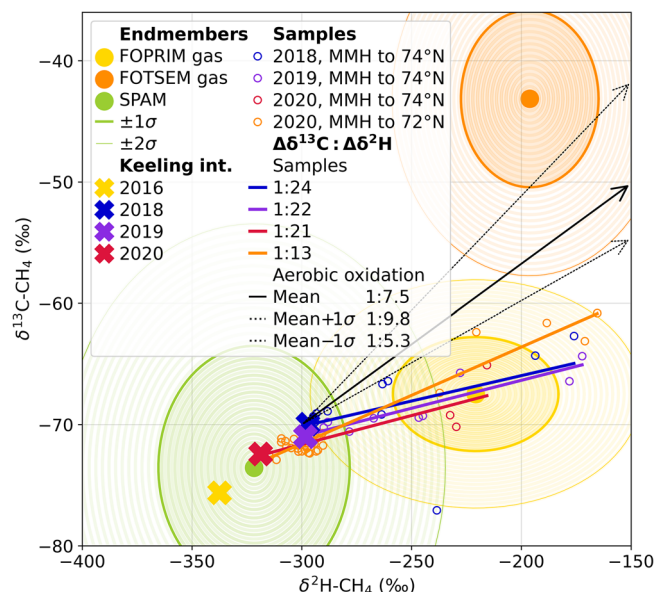


Fig. 5 | Dual-stable isotope plot showing the three main endmembers of old CH₄ in the ESAS, the multi-year Keeling plot-extrapolated source signatures, individual measurements of below pycnocline samples, and putative aerobic CH₄ oxidation fractionation lines. The three main endmembers of old CH₄ (SPAM, FOPRIM gas, and FOTSEM gas) are plotted as mean values (thick dot), $\pm 1\sigma$ (thick line), and $\pm 2\sigma$ (thin line), respectively, in light green, yellow, and orange. The Keeling plot-extrapolated source signatures (Keeling int., Supplementary Table 2) are plotted as crosses in different colors for each year: 2016 (yellow), 2018 (dark blue), 2019 (purple), 2020 (red). The individual dual-stable isotope data of below pycnocline samples are plotted as circles, with different colors for each year and subregion that stretches from the major CH₄ hotspot (MMH) towards either the north (74.0°N) or south (72.0°N) in the inner Laptev Sea: 2018 (dark blue), 2019 (purple), 2020, 73.0–74.0°N (red), 72.0–73.2°N (orange). The linear fits through these measured samples are plotted as straight lines. The putative aerobic CH₄ oxidation fractionation lines ($\pm 1\sigma$, data and references in Supplementary Table 3) are shown as black (and dotted) lines. The sample-derived $\Delta\delta^{13}\text{C}:\Delta\delta^2\text{H}$ lines are all outside the 1σ window of aerobic CH₄ oxidation, indicating that the data are not consistent with this process.

source of the intensive seabed escape of CH₄ in the inner Laptev Sea. The SPAM category includes both freshly produced CH₄ from thawed old OM (>48 ky BP) and CH₄ formed in the past from OM that is now within subsea permafrost, held since formation as free gas or CH₄ hydrate. The triple-isotopic composition of CH₄ cannot be used to differentiate between recently and formerly formed CH₄ when both are formed by the same process from the same OM.

Bayesian MCMC-based source apportionment^{56,57} (Supplementary Methods 1.3.1) was used to constrain the fractional contributions of each of the three identified endmembers while accounting for the isotopic uncertainty of each endmember. As we have no radiocarbon data for 2016 and 2018, we assumed that data for these years would, similar to multi-year consistent stable isotope data, also point towards radiocarbon-depleted CH₄ at the MMH. Therefore, the three considered endmembers for the source apportionment were for each year: SPAM, FOTSEM, and FOPRIM (Supplementary Methods 1.2). The Bayesian MCMC-based results support SPAM as the primary source of CH₄ in the inner Laptev Sea with median fractions of 60–81% SPAM (Fig. 6 and Supplementary Note 8). The source apportionment is consistent for all four study years. Most of the uncertainty in the determined fractional source contributions (Fig. 6) are due to the large variety in the isotopic values found for each endmember category and the inherent large standard deviations on the isotopic values of these endmembers (Supplementary Methods 1.2 and Supplementary Data 1). Small fractions of FOTSEM and FOPRIM gas (medians: 7–16% and 12–24%) may be released simultaneously with SPAM (Fig. 6). These different types of

gases may have mixed while being collectively trapped by the subsea permafrost before their combined release.

Putative release pathways

We deduce that the released SPAM originates predominantly from a pre-formed pool instead of being freshly produced in thawing subsea permafrost. This is supported by the ubiquity of gas bubble fields in the hotspot regions (Fig. 2). Recent modeling studies are inconclusive regarding the contribution of OM degradation in thawing subsea permafrost to the observed CH₄ emissions in the ESAS. One modeling study of subsea permafrost OC stocks proposes low microbial decomposition rates of OM in thawing subsea permafrost ($\sim 0.13 \text{ mmol C m}^{-2} \text{ d}^{-1}$)²⁴. In contrast, another model suggests substantial CH₄ production rates from OM degradation in thawed subsea permafrost ($\sim 11 \text{ mmol CH}_4 \text{ m}^{-2} \text{ d}^{-1}$)⁶³. Extrapolations of incubation experiments using subsea permafrost from the inner Laptev Sea show a minor *in situ* microbial CH₄ production rate of only 0.133 ($\sigma = 0.167$) $\text{mmol CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ ¹¹³. This was the rate estimated for a thawed layer of $14 \pm 7 \text{ m}$ thickness, i.e., the average thaw depth in 2012–2014 of eight subsea permafrost cores from the inner Laptev Sea¹³. Such a limited gas release rate cannot explain the large bubble fluxes observed with the echosounder at the MMH (Fig. 2; Supplementary Note 3 and Supplementary Data 6) nor the observation-based CH₄ ebullition fluxes estimated for this region of 18 (6.2–39) $\text{mmol CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ ¹⁴. We conclude that the released SPAM is dominantly not recently produced but more likely released from a preformed pool of free gas or destabilizing CH₄ hydrate within subsea permafrost. An implication is that the observed CH₄ releases might not necessarily develop gradually but may be more abrupt in nature.

Estimates of CH₄ hydrate abundance on the ESAS vary widely (Supplementary Table 1). Gas hydrates form under specific conditions when CH₄ and H₂O are situated together at a specific ratio, pressure, and temperature²⁵. The pressure and temperature conditions needed to form a CH₄ hydrate restrict CH₄ hydrate formation to the polar regions and deeper (>300 m) oceans^{26,64}. Shallow CH₄ hydrates can be formed at $\sim 200 \text{ m}$ depth in permafrost ($\sim 170\text{--}220 \text{ m}$ at ~ -15 to -5°C)²⁶, and hydrates have been identified in Arctic permafrost^{65–67}. Yet, the gas hydrate stability region may shift upwards/downwards in the presence of other gases, such as carbon dioxide or longer hydrocarbons, and ionic compounds like NaCl²⁶. Furthermore, relict CH₄ hydrates formed by ice sheet-induced pressures during glaciations and subsequently retained outside the hydrate stability zone by self-preservation^{66,68} could be present even shallower in permafrost^{69,70}. However, the lack of ice sheets in the inner Laptev Sea region during the Last Glacial Maximum^{71–73} likely prohibits the growth of hydrates formed due to glacier-exerted pressure. Some modeling studies estimate an upper border of the gas hydrate stability zone in the ESAS at ~ 130 ⁴⁵ and $\sim 75 \text{ m}$ ⁴⁶ during the Pleistocene, while currently at $\sim 200 \text{ m}$ ^{45,46}. Reported subsea permafrost drill cores from the ESAS extend to $\sim 50 \text{ m}$ depth^{4,10}, i.e., not deep enough to verify the presence of hydrates in the ESAS region. All in all, the abundance of CH₄ hydrates on the ESAS and their associated depths is uncertain.

We assess the possibility of gradual thawing and deepening of the subsea permafrost table as a direct cause for the release of preformed SPAM from the thaw front. Preformed SPAM could be present as free gas or as CH₄ hydrate in permafrost, where CH₄ hydrate contains much more CH₄ per unit volume (160–180 times^{74,75}) than free gas. Therefore, we consider destabilizing CH₄ hydrates as the maximum amount of preformed CH₄ that can be released by the gradual deepening of the permafrost table. The deepening of the subsea permafrost table in the coastal inner Laptev Sea was observed to be $\sim 14 \text{ cm y}^{-1}$ ¹¹⁰. To constrain an upper limit for CH₄ release from dissociating CH₄ hydrates, pure CH₄ hydrates were considered. If pure CH₄ hydrates gradually dissociate at $\sim 14 \text{ cm y}^{-1}$ ¹¹⁰, $2700\text{--}3000 \text{ mmol CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ is released^{74,75}. This suggests that CH₄ hydrate dissociation at the thaw front could result in detectable bubble plumes. However, given the depth of the permafrost table at 8–24 m in the shallow coastal inner Laptev Sea^{10,13}, it is unlikely that the thaw front and associated salt intrusion⁷⁶ have yet reached the gas hydrate stability zone^{45,46}. Nevertheless, our study area

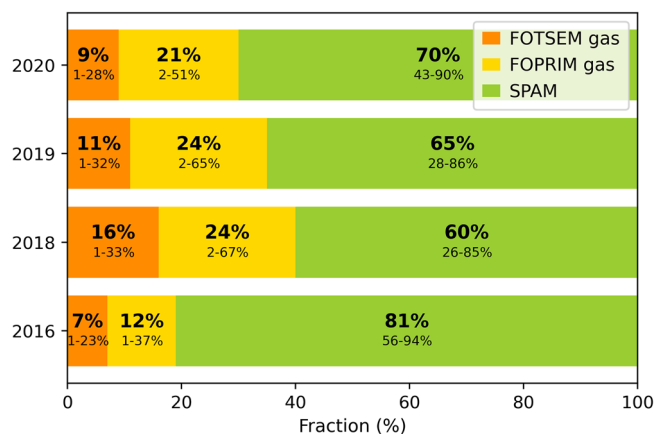


Fig. 6 | Fractional contributions of different subsea sources to the intensive inner Laptev Sea CH₄ releases based on Bayesian MCMC isotopic source apportionment. The median fraction and 90% credible intervals for each source category and year are shown. The colors in the bar plot indicate the different subsea sources: SPAM (light green), FOPRIM gas (yellow), and FOTSEM gas (orange). The probability density functions of the fractional source contributions of each source are shown in Supplementary Note 8.

was inundated earlier than the locations with permafrost table data^{7,8,10,13}. Due to longer exposure to heat and salinity from overlying seawater, our study area may therefore exhibit a slightly deeper permafrost table and thaw front. Furthermore, the combination of geothermal heat and warming induced by the overlying seawater decreases the gas hydrate stability zone below the permafrost table^{45,46}, possibly resulting in hydrate destabilization below the thaw front⁷⁷. However, when considering permafrost as an impermeable layer for gas flow, the migration of CH₄ released from such destabilizing CH₄ hydrates below the thaw front would require thawed-through zones that extend much deeper than the current thaw front, such as taliks, as discussed next.

Taliks of different origin may act as conduits for upward-moving CH₄. The inner Laptev Sea is tectonically active, and the study area is located in the Ust'-Lensky rift basin, which contains many faults^{29,34,35,78}. Such faults and the associated taliks from penetrating geothermal heat^{46,79} are a potential pathway for the release of SPAM. Other types of taliks, from paleo-river channels or submerged thaw lakes, have also been reported to be present in the inner Laptev Sea and could also act as vertical conduits for the release of preformed CH₄^{10,29,32,80}. Taliks formed by paleo-river channels have been modeled as being able to explain gas venting from dissociating hydrate deposits³³. Warming from both the top and the bottom after submergence was modeled to increase talik width, resulting in gas hydrate dissociation near the talik edges, driving gas transport to the water column³³. Similarly, open taliks, acting as gas migration pathways for preformed subsea-permafrost trapped CH₄, have been modeled to be present over large areas in the inner Laptev Sea⁴. Furthermore, sub-bottom profile images for the coastal Laptev Sea showed blanked areas interpreted as propagating gas plumes co-located with a submerged lake basin¹⁰. The overlying bubble plumes in the water column pointed to gas propagation through a submerged thaw lake talik, and it was hypothesized that such submerged thaw lake taliks develop further downwards after inundation^{10,29,81}. Further mechanisms for talik formation have also been observed. On the Beaufort Sea shelf, gas flares were observed to be associated with pingo-like features⁸². These were interpreted as being formed by upward sediment extrusion due to sub-permafrost gas pressure build-up from gas hydrate decomposition⁸², an example of gas release and talik formation driven by geothermal heat from below. In summary, existing and developing taliks of different origins could explain the observed release of SPAM in the inner Laptev Sea.

At the major CH₄ hotspot in our study area, we have identified SPAM as the primary source, while fractions of FOPRIM and FOTSEM gases are also suggested (Fig. 6 and Supplementary Note 8). The release of CH₄ along

taliks from subsea permafrost-trapped gas, potentially from dissociating CH₄ hydrates, aligns with these observations as SPAM could have mixed with FOPRIM and FOTSEM gases while being trapped. Furthermore, the demonstrated presence of taliks in the inner Laptev Sea and the propagation of gas plumes through the seabed^{2,10,83} make CH₄ release along these features plausible. Moreover, local ebullition hotspots at many locations in the inner Laptev Sea align with developing taliks releasing preformed CH₄. Therefore, we believe that the leakage of subsea permafrost-associated CH₄ along existing and/or developing taliks is the most likely explanation for the observed intensive CH₄ release at the major CH₄ hotspot in the inner Laptev Sea. Taken all together, the observed CH₄ is released from a preformed CH₄ pool associated with degrading subsea permafrost.

The fate of methane after release from the seabed

The old microbial subsea permafrost-associated CH₄, released at the major CH₄ hotspot in the inner Laptev Sea, is observed to mix with a younger microbial, likely partly oxidized, CH₄ background in the water column. All samples below the pycnocline with dual-stable isotope data lie within stable isotope regimes of CH₄ formed by primary microbial pathways^{53,54} (Fig. 5). This aligns with the proposed scenario in which CH₄ is predominantly released from an old microbial CH₄ source, mixing into a younger (Fig. 3) yet microbial CH₄ background. The observed trend of enrichment in the stable isotopes with decreasing concentration is consistent with a partly oxidized CH₄ background pool, considering the enriched stable isotopic composition of the background material^{53,54} (Fig. 4). The different slopes of the Keeling plots to either the north (74°N) or south (72°N) of the MMH (Fig. 4) could point to mixing with background material having different concentrations and/or more/less isotopically enriched CH₄. The isotopic enrichment could also partly originate from aerobic oxidation of the CH₄ released at the MMH. However, the stable isotope Keeling plots show strong linear relationships, and the $\delta^2\text{H}_{\text{CH}_4}$ data manifest as a cloud around -300‰ across a range of CH₄ concentrations at the MMH in 2020 (Fig. 4). This suggests that CH₄ oxidation is not an important cause of the varying CH₄ concentrations. Instead, CH₄ from the same source appears to be released/mixed into the water column to different extents, possibly through bubble dissolution, resulting in the observed variation in CH₄ concentration.

Additional lines of evidence support that there is little microbial CH₄ oxidation of the primary source in the water column. The dual-stable isotope plot (Fig. 5) shows a range of literature-based dual-stable isotope fractionation lines derived from microbial incubation experiments (Supplementary Table 3). These lines were derived by computing the shift in $\delta^{13}\text{C}_{\text{CH}_4}$ for the CH₄ fraction remaining corresponding to a $\delta^2\text{H}_{\text{CH}_4}$ shift from -300‰ to -150‰ of the source material. The resulting theoretical range of the dual-stable isotope fractionation lines for microbial aerobic oxidation in the water column, i.e., a $\Delta\delta^{13}\text{C}:\Delta\delta^2\text{H}$ ratio of 7.5 ± 2.2 ($\pm 1\sigma$), does not align with the fitted $\Delta\delta^{13}\text{C}:\Delta\delta^2\text{H}$ lines through the measured dual-stable isotope data, as these were much larger, ranging from 13 to 24 (Fig. 5). This indicates that the change in stable isotope composition in our seawater samples does not reflect CH₄ oxidation in the water column, but is more likely caused by mixing with a background pool. Importantly, any dilution of the primary source material and additional source addition, for example, by bubble dissolution, would keep the observed stable isotope data along the mixing line. Finally, the bottom, mid-column, and surface waters at the MMH show CH₄ of similar stable isotopic composition (Fig. 7), which again aligns with minor CH₄ oxidation in the water column in this region.

Based on the above, it is likely that a predominant fraction of the subsea permafrost-associated CH₄ released at the major CH₄ hotspot reaches the surface water and atmosphere in this shallow ocean region. Similar source signatures were identified with Keeling plots below and above the pycnocline (Fig. 4 and Supplementary Table 2), identifying the same primary CH₄ source in both surface and bottom waters. Moreover, south of the hotspot, more negative $\delta^2\text{H}_{\text{CH}_4}$ values were found in the surface waters compared to the bottom waters (Fig. 7). This could point to CH₄ being spread along the

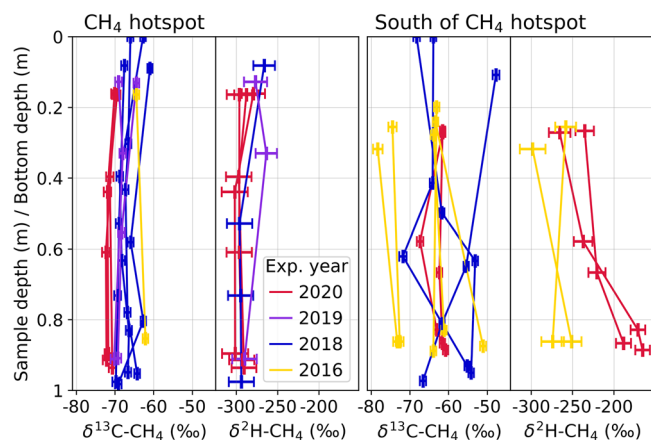


Fig. 7 | Water column stable isotope data at the major CH₄ hotspot and south of this hotspot in the inner Laptev Sea. Dual-stable isotope data are plotted for stations with water column samples that include at least one bottom water sample (0–3 m from the bottom) and one surface water sample (0–5 m from the water surface). The expedition years (exp. year) are marked with different colors: 2016 (yellow), 2018 (dark blue), 2019 (purple), and 2020 (red). Error bars represent ± 2 standard uncertainty. On the y-axis, a ratio of 1 indicates the bottom level, and a ratio of 0 the water surface. The complete sample lists and stable isotope data are in Supplementary Data 2–5.

surface away from the MMH, leading to a somewhat lower $\delta^2\text{H}_{\text{CH}_4}$ value in the surface water south of the MMH. Furthermore, the rising bubbles likely transport CH₄ directly to the atmosphere, considering the shallow water column⁸⁴. Even if CH₄ dissolves partly from rising bubbles through the shallow water column at the MMH, resulting in similar CH₄ stable isotope signals along the water column up to the surface water (Fig. 7), the remaining CH₄ in such bubbles reaching the surface water will be directly emitted to the atmosphere.

Lower CH₄ concentrations but similar stable isotope composition in the surface water compared to sub-pycnocline water at the MMH could point to several processes. At the MMH, CH₄ concentrations were smaller in the surface water than in the bottom water (Fig. 2; Supplementary Note 2 and Supplementary Data 2–5). The dissolved CH₄ in the surface water was likely diluted through water mixing, reducing its observed concentration. Furthermore, CH₄ can diffuse from the water column into the atmosphere, reducing its concentration. An effective way to rapidly reduce CH₄ concentrations in surface waters is by intensive CH₄ release (ventilation) due to storm events, which has been observationally demonstrated for this region⁴. Both slow diffusion and rapid CH₄ venting result in the release of SPAM from the surface water into the atmosphere. The relatively invariant dual-stable isotope composition along the water column at the MMH (Fig. 7) and deduced CH₄ release to the atmosphere contrast with findings for the Beaufort Shelf, where it was suggested that CH₄ oxidation and dispersion strongly limit the emission of ancient CH₄ into the atmosphere⁵⁵. All in all, this study suggests the release of old microbial preformed CH₄ from deep seabed sources in degrading permafrost that reaches the surface water and atmosphere in the inner Laptev Sea region.

Implications

Multi-year triple-isotopic analysis of seawater CH₄ from the inner Laptev Sea region with intensive sedimentary CH₄ emissions indicates the release of predominantly old, microbial CH₄ formerly trapped in subsea permafrost. Based on the ebullitive character and the geological setting, it is hypothesized that (developing) taliks facilitate the seepage of this preformed CH₄, possibly from destabilizing CH₄ hydrates. Importantly, our results point towards the release of preformed CH₄ from degrading subsea permafrost, hereby providing insights on a vulnerable CH₄ compartment in the context of climate warming.

The major finding of old, microbial, and subsea permafrost-associated CH₄ as the primary source in the inner Laptev Sea contrasts with thermogenic CH₄ released in the outer Laptev Sea. This indicates that multiple CH₄ pools contribute to the observations of elevated CH₄ concentrations on the extensive ESAS, the largest shelf region of the world. This implies that multiple sources and release pathways must be considered in estimates of current and future CH₄ releases from the ESAS system. Furthermore, considering the size of the subsea permafrost-associated CH₄ pool and its vulnerability, this may be an intensifying natural CH₄ release process that may be one part of the poorly understood current increase in global atmospheric CH₄, where enhanced natural releases of biogenic CH₄ are suggested^{85,86}. The identified CH₄ release from subsea permafrost, a little-studied yet major earth system compartment, is exacerbated by the inefficient removal through microbial oxidation in the water column, likely yielding the majority of the CH₄ being also released into the atmosphere. The observed release of formerly subsea-permafrost-trapped CH₄ in the inner Laptev Sea is coupled to subsea permafrost destabilization, a process that is expected to continue with global warming and needs further consideration.

Methods

Expeditions and sampling stations

This study is based on four extensive expeditions to the distant ESAS. In 2016, the R/V Akademik M.A. Lavrentyev traversed the inner Laptev Sea at the end of September while in October 2018, 2019, and 2020, the R/V Akademik Mstislav Keldysh (AMK) crossed the inner Laptev Sea (Fig. 2). Sampling was focused on regions where high CH₄ concentrations had been previously observed in order to address the source(s) of these massive sedimentary releases. Within these regions, geophysical soundings were used to detect gas seeps that could indicate strong CH₄ release by ebullition. Most sampling took place close to seep sites, while some samples were retrieved at background locations with lower CH₄ concentrations for comparison (sample details in Supplementary Data 2–5).

Seep detection

The seeps were detected with a scientific 200 kHz single-beam echosounder, Simrad EK15. The beam swath was 26°, resulting in a swath diameter of ~20 m and a sounding area of ~300 m² at ~20 m water depth. The acoustic survey was conducted at ~11 km h⁻¹. The presence of rising bubbles was identified by observing an increased acoustic signal intensity in the water column (Supplementary Note 3). This type of echosounder data was generated for the expeditions in 2018, 2019, and 2020. This seep detection technique has been described elsewhere^{4,5,87}.

Procedures for sampling seawater methane for concentration and isotopic analyses

Seawater samples for CH₄ concentration and isotope analyses were retrieved with a Niskin Rosette, deployed from the research vessel (procedures in Supplementary Methods 1.1.1). The samples for CH₄ concentration and stable isotope analysis were taken at several water depths. In 2020, additional samples for stable isotope analysis were collected at the three main CH₄ hotspot locations in the form of a set of horizontally spaced near-bottom water samples. Samples for radiocarbon content were collected from bottom water at locations where high CH₄ concentrations were found.

Methane concentration, stable isotope, and radiocarbon analysis

Methane concentrations were measured onboard after headspace equilibration using gas chromatography flame-ionization detection (GC-FID, Supplementary Methods 1.1.2). Gas samples, retrieved by headspace equilibration onshore, were analyzed for CH₄ stable isotope composition using gas chromatography continuous-flow isotope-ratio mass spectrometry (GC-CF-IRMS) and two standards with different isotopic composition (Supplementary Methods 1.1.3). Upon entering the GC-CF-IRMS, CH₄ in the gas samples was concentrated in a preconcentration device resembling a

former setup⁸⁸ but with some modifications (Supplementary Methods 1.1.3). Samples for radiocarbon analysis were sampled and prepared for analysis by accelerator mass spectrometry following a published method⁸⁹ (summary in Supplementary Methods 1.1.4).

Isotopic endmembers of putative subsea methane sources

Three main endmembers were considered for the old (>48 ky BP) CH₄ that was found. These are: subsea permafrost-associated methane (SPAM), fossil primary microbial (FOPRIM) gas, and fossil thermogenic and secondary microbial (FOTSEM) gas (Fig. 1). Full explanations of the endmembers, the isotopic endmember database, and the resulting constraints on the isotopic source endmember values can be found in Supplementary Methods 1.2 and Supplementary Data 1. Isotopic data from 158 articles were included in the isotopic endmember database. In total, 642, 484, and 2394 isotopic samples were incorporated in the database to construct the isotopic endmember values for, respectively, SPAM, FOPRIM gas, and FOTSEM gas.

In short, SPAM covers all forms of microbial CH₄ (free gas or hydrate) formed from OM in the upper subsea permafrost region before/during freezing and formed upon subsea permafrost thawing. Importantly, this category thus also encompasses CH₄ formed from relatively fresh OM accumulated during the past several million years and trapped by permafrost after formation, i.e., a preformed CH₄ pool. As SPAM represents primary microbially formed CH₄, the constrained dual-stable isotope values of this endmember are low (Supplementary Methods 1.2.4 and Supplementary Data 1). The endmember SPAM is situated in the subsea permafrost layer (Fig. 1).

The natural gases FOTSEM and FOPRIM are fossil gases formed below permafrost by either catagenesis or primary/secondary microbial processes. The endmember FOPRIM encompasses primary microbial gas formed by OM degradation at temperatures up to 60–80 °C^{90–92}, using older OM than SPAM that can be up to several hundred million years old. It is typically formed up to depths of ~1–3 km^{91–93}. The endmember FOTSEM encompasses CH₄ formed by catagenesis in even deeper sedimentary basins at a pressure of 300–1500 bar and a temperature of 50–200 °C^{90,94}. Furthermore, it encompasses secondary microbial CH₄ formed by the biodegradation of thermogenic petroleum, especially oil, in relatively cold (<70–90 °C) reservoirs⁹⁵. The constrained stable carbon isotope value of the endmember FOPRIM gas is much more negative than that of FOTSEM gas due to the primary microbial origin of FOPRIM gas (Supplementary Methods 1.2.1–1.2.3 and Supplementary Data 1). Both FOPRIM and FOTSEM gases can be present at the location of formation, accumulate in another reservoir, or get trapped as CH₄ hydrate or free gas in or below subsea permafrost. While isotopic compositions alone do not allow for further distinction between these compartments, the endmembers FOPRIM and FOTSEM effectively represent ancient preformed CH₄ pools. The endmembers FOPRIM and FOTSEM gases are indicated with yellow and red, respectively, in Fig. 1.

The constrained isotopic endmember values for $\delta^{13}\text{C}_{\text{CH}_4}$ are $-43 \pm 7\text{‰}$, $-67 \pm 5\text{‰}$, and $-74 \pm 10\text{‰}$ for, respectively, FOTSEM, FOPRIM, and SPAM. The constrained isotopic endmember values for $\delta^2\text{H}_{\text{CH}_4}$ are $-196 \pm 31\text{‰}$, $-221 \pm 38\text{‰}$, and $-322 \pm 44\text{‰}$ for, respectively, FOTSEM, FOPRIM, and SPAM (database containing individual isotopic data in Supplementary Data 1).

Statistical source apportionment of the released methane

The isotopic measurements of dissolved CH₄ were extrapolated to the source signatures of the CH₄ being released to the water column using Keeling plots. To obtain the Keeling plots, a linear regression model was fitted to the measured isotopic values versus the inverse CH₄ concentration using the least squares method. Keeling plots^{3,96} are applicable for source determination when a distinct single source or well-mixed group of sources mixes to different extents into relatively constant background material. If a mixing scenario is valid, the y-intercept of the Keeling plot gives the isotopic signature of the source(s). A prerequisite for using Keeling plots is that fractionating processes, like CH₄ oxidation, do not dominate the observed

isotopic changes in the mixing medium, which is seawater in this case. In our study, the Keeling plot-extrapolated values (y-intercepts) were combined with the constrained endmember data in a Bayesian MCMC approach^{56,57} to derive the probability density functions of the fractions of each source category. A Bayesian MCMC approach was used to account for the isotopic uncertainty of the identified endmembers and to obtain the probability density functions of the source fractions, thereby quantifying the uncertainties associated with these results. A complete description of this procedure can be found in Supplementary Methods 1.3.

Data availability

All data used in this study (Supplementary Data 1–10) are publicly available at the Bolin Centre for Climate Research database (<https://doi.org/10.17043/brussee-2026-methane-1>).

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Author contributions

M.B.: conceptualization, laboratory work, formal analysis, writing original draft, review and editing. H.H.: conceptualization, laboratory work, review and editing. B.W., D.K., D.C., and A.K.: fieldwork, review and editing. N.S.: review and editing. I.S., Ö.G.: conceptualization, fieldwork, review and editing.

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